THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 03

Full Syllabus (Chapters 1–14) | Class 10 Mathematics | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** 323	Distinct primes
2	**(b)** 216	12×18
3	**(c)** 4	Zeroes = x-intercepts
4	**(b)** $0, 2, -2$	$x(x-2)(x+2)$
5	**(c)** coincident	All ratios $\frac{1}{2}$
6	**(a)** Student A is correct	Put $x = 0$; B found the x-intercept
7	**(c)** $\frac{3}{2}$	$(2x-3)(x+2)$
8	**(b)** Incorrect — -6 and -5 also work	$x^2 + x - 30 = 0 \Rightarrow x = 5$ or -6
9	**(d)** $3, 5, 7$	$n = 1, 2, 3$
10	**(c)** 31	$\frac{90}{3} + 1$
11	**(a)** 6 cm	$\frac{3}{DB} = \frac{1}{2}$
12	**(a)** one angle equal and the sides including it proportional	Criterion
13	**(a)** $\frac{5}{3}$	Reciprocal
14	**(a)** 130°	Supplementary
15	**(a)** Correct	Half the circle
16	**(d)** 346.5 cm²	$\frac{1}{4} \times 1386$
17	**(d)** $4\pi r^2$	It is a sphere
18	**(c)** Incorrect — it is the class with the highest frequency	Definition
19	**(b)** 25	Larger end
20	**(d)** $\frac{1}{8}$	1 of 8

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	$24 = 2^3 \times 3,\ 36 = 2^2 \times 3^2$; HCF $= 12$, LCM $= 72$; $12 \times 72 = 864 = 24 \times 36$ ✓	1+1
22	$x^2 - 0x + \sqrt{5} = x^2 + \sqrt{5}$	2
23	$16x^2 - 8x + 1 = 0 \Rightarrow (4x-1)^2 = 0 \Rightarrow \frac{1}{4}, \frac{1}{4}$	1+1
24	$\angle C = 65^\circ \Rightarrow \angle B = 180^\circ - 97^\circ = 83^\circ$	1+1
25	(i) 13 (ii) 5	1+1
26	$\theta = 30^\circ \Rightarrow \frac{1}{4} - \frac{3}{4} = -\frac{1}{2}$	1+1
27	$1000\sqrt{3} \approx 1732$ m	2
28	$AB + CD = BC + AD \Rightarrow AD = 8$ cm	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Points $(0,2), (4,0)$; $(0,3), (6,0)$. Lines are parallel ⇒ inconsistent (no solution)	2+1
30	$n = 30$; cf 3, 10 ⇒ $150 + \frac{15-10}{10} \times 5 = 152.5$ cm	1+1+1
31	$3a_{19} = 225 \Rightarrow a + 18d = 75$; $3a_{36} = 429 \Rightarrow a + 35d = 143 \Rightarrow d = 4,\ a = 3 \Rightarrow 3, 7, 11, \ldots$	1+1+1
32	$AS = 18 = AR \Rightarrow BR = 29 - 18 = 11$; OQBR is a square ⇒ $r = 11$ cm	1+1+1
33	$113.04 - 72 = 41.04$ cm²	1+1+1
34	Blue 18, green 24 ⇒ white 12	1+1+1
35	$60\sqrt{3} + 60 = 60(\sqrt{3}+1) \approx 163.92$ m	1+1+1

	OR: $h = x\sqrt{3} = \frac{x+30}{\sqrt{3}} \Rightarrow x = 15$ m; $h = 15\sqrt{3} \approx 25.98$ m	1+1+1
36	$a = 50, h = 10$; $\Sigma fu = -15$ ⇒ $50 - 5 = 45$ years	1+1+1
	OR: Modal class 10–20: $10 + \frac{14-7}{28-7-13} \times 10 = 18.75$	1+1+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) 10 (ii) $\left(\frac{17}{5}, \frac{26}{5}\right) = (3.4, 5.2)$ (iii) $F = (13, 18)$ (iv) 20	1+2+1+1
	OR: (i) 10 (ii) $\left(\frac{28}{5}, \frac{39}{5}\right)$ (iii) $(x-2)^2 + 9 = (x-8)^2 + 121 \Rightarrow x = \frac{43}{3}$ ⇒ $\left(\frac{43}{3}, 0\right)$	1+2+2
38	$x^2 - y^2 = a^2(\sec^2\theta - \tan^2\theta) + b^2(\tan^2\theta - \sec^2\theta) = a^2 - b^2$ (cross terms cancel)	2+2+1
	OR: Write $1 = \sec^2 A - \tan^2 A$ in the numerator: $\frac{(\sec A+\tan A)(1-\sec A+\tan A)}{\tan A-\sec A+1} = \sec A + \tan A = \frac{1+\sin A}{\cos A}$	2+2+1
39	(i) $10\pi + \frac{2}{3}\pi = \frac{32}{3}\pi \approx 33.52$ cm³ (ii) $20\pi + 2\pi = 22\pi \approx 69.14$ cm² (iii) $1000 \div 33.52 \approx 29.8$ ⇒ 29	2+2+1
	OR: (i) 753.6 cm³ (ii) 133.97 cm³ (iii) 887.57 cm³ ≈ 0.89 L (iv) 376.8 cm²	1+1+2+1
40	$x + y = 80$; $46 = 40 + \frac{115-(42+x)}{65} \times 10 \Rightarrow x = 34,\ y = 46$	2+2+1
	OR: $\frac{940+35p}{44+p} = 25 \Rightarrow p = 16$ (3). $n = 60$; cf 5, 23 ⇒ $20 + \frac{30-23}{15} \times 10 = 24.67$ (2)	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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